

PROFILE

Fourth-year B.E. Chemical Engineering student at Kathmandu University, expected to graduate in 2026. Interested in chemical reactions and sustainable engineering, with project experience in process design and simulation, materials science, geothermal energy, and green chemistry. Brings analytical problem-solving, scientific programming, laboratory project work, and teaching and mentoring experience.

CURRENT EXPERIENCE

CG Brewery

Nawalparasi, Nepal

INTERN · CURRENT

SELECTED PROJECTS

CO₂ Utilization for Green Methanol

Fourth-Year Plant Design · KU

Team conceptual design and Aspen Plus V15 simulation of a power-to-methanol process in Nepal. The integrated flowsheet covers MEA-based CO₂ capture from cement-kiln flue gas, alkaline water electrolysis for green hydrogen, reverse water-gas shift, a methanol synthesis loop with recycle, and two-stage distillation. Scope also includes process flow and piping and instrumentation diagrams, mass and energy balances, equipment and instrumentation specification, and techno-economic assessment. The study also considered cost sensitivities and process-improvement options, including hydrogen recovery from purge gas, staged RWGS with water removal, and heat integration.

MSWI Fly Ash Utilization in Cement Clinker Production

2025

Departmental research project examining the feasibility and techno-economic case for incorporating municipal solid-waste incineration fly ash into cement clinker production. Considered environmental implications and the circular-economy potential of using an industrial waste stream as a clinker substitute.

Geothermal Energy — Air-Based Ground Source Heating

2024 · WEIR Lab

Designed an air-based ground-source heating system at WEIR Lab, Kathmandu University. Carried out heat-transfer analysis and optimisation, then evaluated thermal performance and the economic viability of ground-source heat-pump technology.

Photocatalytic Decomposition of Ethylene

2024

Departmental semester project investigating catalytic pathways for ethylene removal to support longer post-harvest agricultural shelf life. Planned controlled experiments and considered catalyst-performance parameters for optimisation.

Biodegradable Plastic Film from Potato Starch

2023

Synthesised bio-based polymer films from potato starch, characterised mechanical and thermal properties, and evaluated biodegradability against conventional plastics using green-chemistry principles.

EDUCATION

B.E. Chemical Engineering	2022–present
Kathmandu University · Fourth year · Expected graduation 2026	
School Leaving Certificate (+2)	2022
KU High School	
Secondary Education Examination	2019
Siddhartha School and College	

LEADERSHIP & COMMUNITY

Electronics Camp Organiser	2024
Independent initiative	
Delivered a three-month curriculum in embedded systems, sensors, and actuators. Mentored school students through C/C++ projects including automated lighting and obstacle-avoiding robots.	
Member · KUCONC	2022–present
Kathmandu University Circle of Noble Chemineers	
Assistant Teacher · Mathematics & Science	2017–2020
Siddhartha Secondary School	

AWARDS & CERTIFICATES

Japanese Language Scholarship	2024
FORDAYS Co. Ltd., Japan · USD \$2,000 award and industry visit	
Python for Web App Development	2020
Course certification	
Data Management for Cyber Security	2020
Course certification	

References available upon request

TECHNICAL SKILLS

Programming & scientific computing
Python, NumPy, Pandas, Matplotlib, Scikit-learn, C, C++
Process engineering & simulation
Aspen Plus, plant design, process flow and P&ID, reactor design
CAD, laboratory & materials
AutoCAD, SolidWorks, Fusion 360, materials characterisation
Methods and application
Process simulation, heat-transfer analysis, experimental design, techno-economic assessment, environmental evaluation

LANGUAGES

English · Fluent
Nepali · Native
Hindi · Conversational
Japanese · JLPT N5

AREAS OF INTEREST

Chemical reactions and process engineering
Sustainable engineering and circular materials
Renewable energy and green chemistry
Materials science and computational simulation